

YASH P SHETTY

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Career Objective

Motivated Electronics and Communication Engineering student with knowledge in digital systems, signal processing, and embedded technologies. Interested in IoT, VLSI, and automation with a goal of applying technical skills to real-world engineering problems and gaining industry experience.

Education

NMAM Institute of Technology, Nitte 2023 – 2027
B.Tech in Electronics and Communication Engineering

SDPT PU College, Kateel, Mangalore 2021 – 2023

St. Joseph's High School, Bajpe, Mangalore 2021

Internship

Mangalore Refinery and Petrochemical Ltd (MRPL) June 1 – June 16, 2025
Instrumentation Department Trainee

- Exposure to industrial instrumentation systems, control loops, and process measurement devices.
- Observed maintenance and calibration of field instruments such as transmitters and valves.
- Assisted engineers with operational reporting and safety procedures.

Invent Health

Feb 25 – Present, 2026

- Developing AI-based healthcare applications using Python, LLMs, and RAG.
- Integrating Hugging Face models for medical information retrieval and conversational AI.
- Building and optimizing NLP pipelines for accurate healthcare assistance.

Projects

Multi-Modal AI-Based Silent and Audio Distress Detection Framework Using Edge-IoT Communication for Intelligent Emergency Assistance (C,Python,AI, IoT, Edge Computing)

- Developed a multi-modal AI framework to detect emergency distress situations using both silent gestures and audio-based signals.
- Implemented machine learning models in Python for real-time distress recognition and classification.
- Designed an Edge-IoT communication architecture to enable low-latency emergency alerts and intelligent assistance.
- Integrated sensor data processing, event detection, and automated notification mechanisms to improve response time in critical situations.

Brain Tumour Detection using MRI Image (Python,Deep Learning, Computer Vision)

- Developed an AI-based medical imaging system for automated brain tumor detection from MRI scans
- Applied image preprocessing, feature extraction, and deep learning techniques to improve diagnostic accuracy.
- Trained and evaluated convolutional neural network (CNN) models for tumor classification and detection.
- Built a prediction pipeline capable of assisting healthcare professionals in early diagnosis and treatment planning.

Code-Converter Application (Python,LLMs, Hugging Face, NLP)

- Developed an LLM-powered code conversion application capable of translating source code between five programming languages.
- Integrated Hugging Face transformer models to generate accurate and context-aware code transformations.
- Designed a user-friendly interface that streamlines cross-language software development and learning

Medical AI Assistant Application (Python, LLMs, RAG, Hugging Face)

- Built an intelligent Medical AI Assistant leveraging Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG).
- Integrated medical knowledge retrieval systems to provide accurate, context-aware healthcare information.
- Developed a document indexing and semantic search pipeline for efficient access to medical resources.
- Implemented conversational AI capabilities using Hugging Face models to deliver reliable and user-friendly medical assistance.

Nutrient Calculator (Python)

- Created a Nutrient Calculator application using Python.
- Functionality includes determining and tracking daily nutritional intake. based on user input.
- Emphasized algorithmic design and data handling.

Technical Skills

Programming:

- C
- C++
- Python

Engineering Tools:

- MATLAB
- Verilog
- SystemVerilog

Domain Focus:

- Signal Processing
- Digital Design
- Instrumentation

Soft Skills

- Time Management
- Teamwork
- Problem Solving

Certifications

- AINNOVATION 2025: Applied AI Learning (Microsoft)
- AINNOVATION 2025: Microsoft Azure Learning
- AINNOVATION 2025: Microsoft AI Learning
- MATLAB for Signal Processing (On-Ramp)

Languages

- English (Fluent)
- Kannada (Native/Fluent)
- Tulu (Native/Fluent)